

Lecture 8 Tutorial — Non-parametric Learning

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Q1. Non-parametric Modeling (Fill in the blanks)

Complete the statements.

Non-parametric models can be used with arbitrary distributions and without assuming that the forms of the underlying _____ are known.

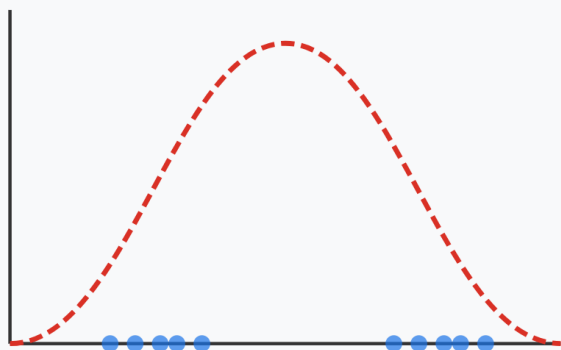
Non-parametric models are suitable for _____ distributions, which are common in practice.

Non-Parametric Modeling

- Non-parametric models can be used with arbitrary distributions and without the assumption that the forms of the underlying densities are known
- Moreover, they can be used with multimodal distributions which are much more common in practice than unimodal distributions.

Parametric vs. Non-parametric Models (Fitting Multimodal Data)

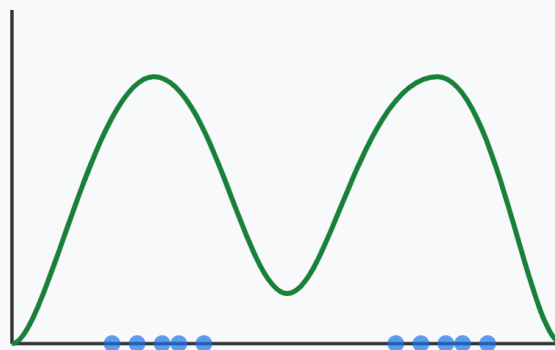
Parametric Model (e.g., Single Gaussian)



Assumes fixed distribution form, fails to fit two peaks

Result: Underfitting

Non-parametric Model (e.g., KDE)



Data-driven without fixed assumption, captures bimodal shape

Result: Accurate Fit

• Real Data Points (Multimodal)
— Model Fitted Curve

1. underlying densities
2. multimodal

Q2. Pros and Cons (True / False)

Write T (True) or F (False).

1. Non-parametric methods require no distributional assumptions ahead of time.
2. With enough samples, non-parametric density estimation can converge to an arbitrarily complicated target density.
3. The number of samples needed typically grows linearly with feature dimension.
4. Non-parametric methods can be sensitive to the choice of window size.

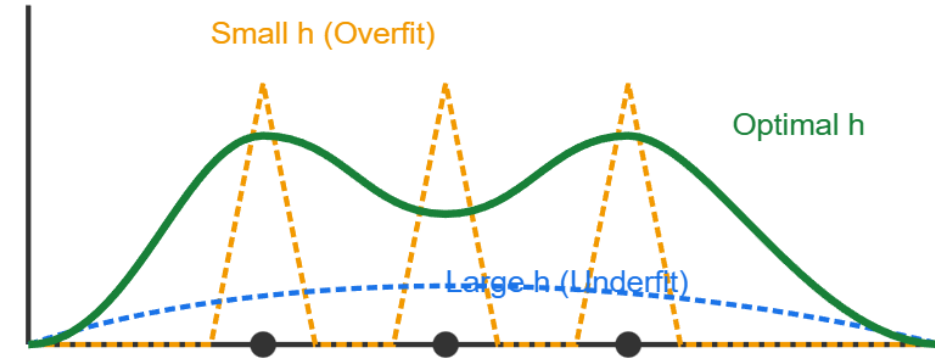
Q1 & Q2: No Assumptions & Arbitrary Complexity

Can fit any complex target density given enough samples.



Q4: Sensitive to Window Size (h)

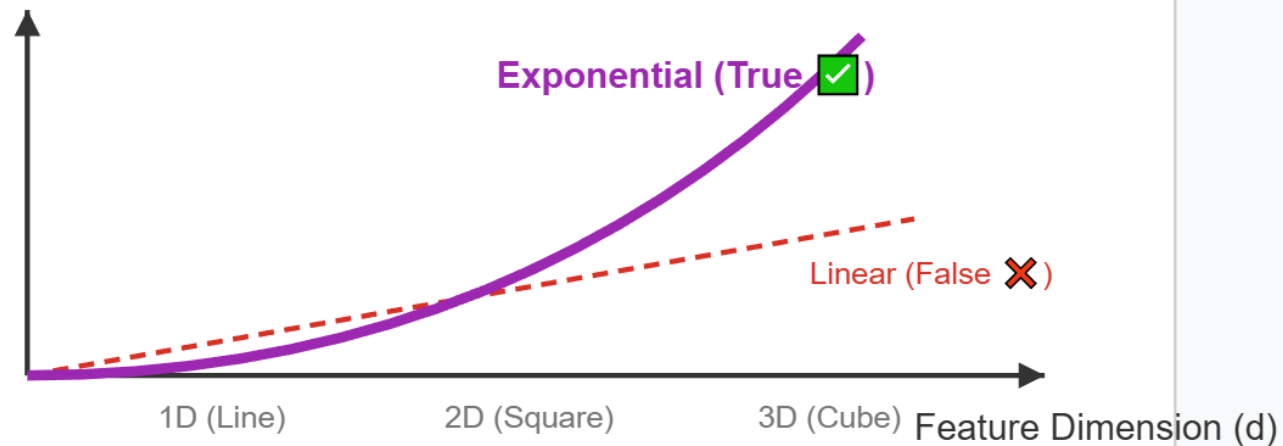
Bandwidth choice drastically changes the density estimate.



Required Samples (N)

Q3: Curse of Dimensionality (Exponential Growth)

Samples needed grow EXPONENTIALLY (not linearly) with dimensions.



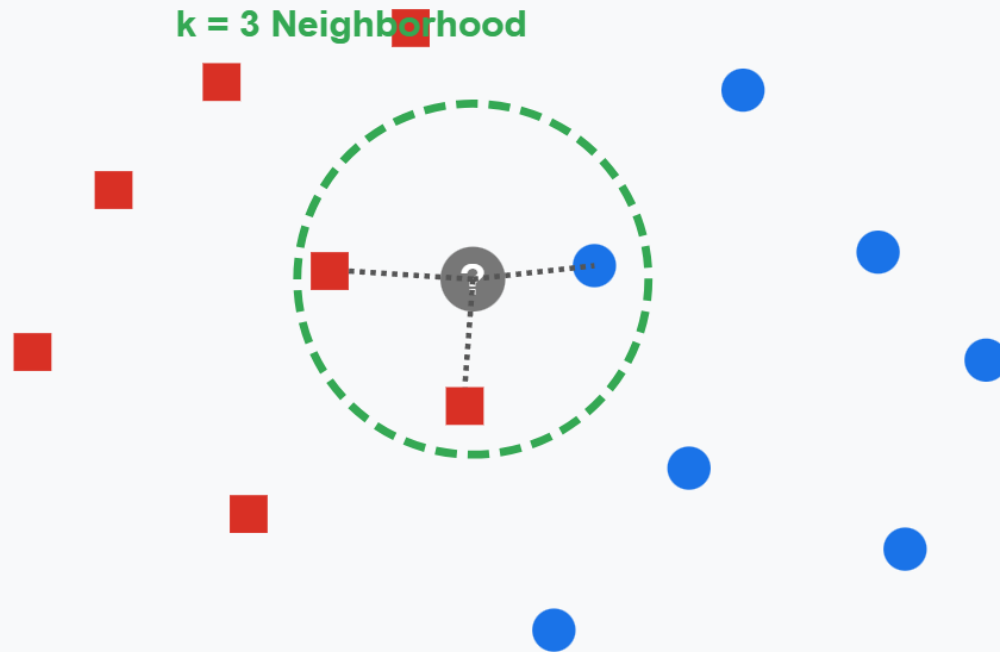
Q3. kNN Classification Rule (Multiple Choice)

In kNN classification, the predicted label of a new point is determined by:

- A. the label of the farthest neighbour
- B. the average of neighbour labels
- C. majority vote among the nearest neighbours' labels
- D. maximizing the likelihood of a Gaussian model

k-Nearest Neighbors (kNN) Classification Rule

The predicted label is determined by a Majority Vote



How to Classify "?"

Step 1: Select k (e.g., $k = 3$).

Step 2: Find 3 closest neighbors.

Step 3: Majority Vote

■ 2 Red Squares

● 1 Blue Circle

Result: "?" is predicted as RED

C. majority vote among the nearest neighbours' labels

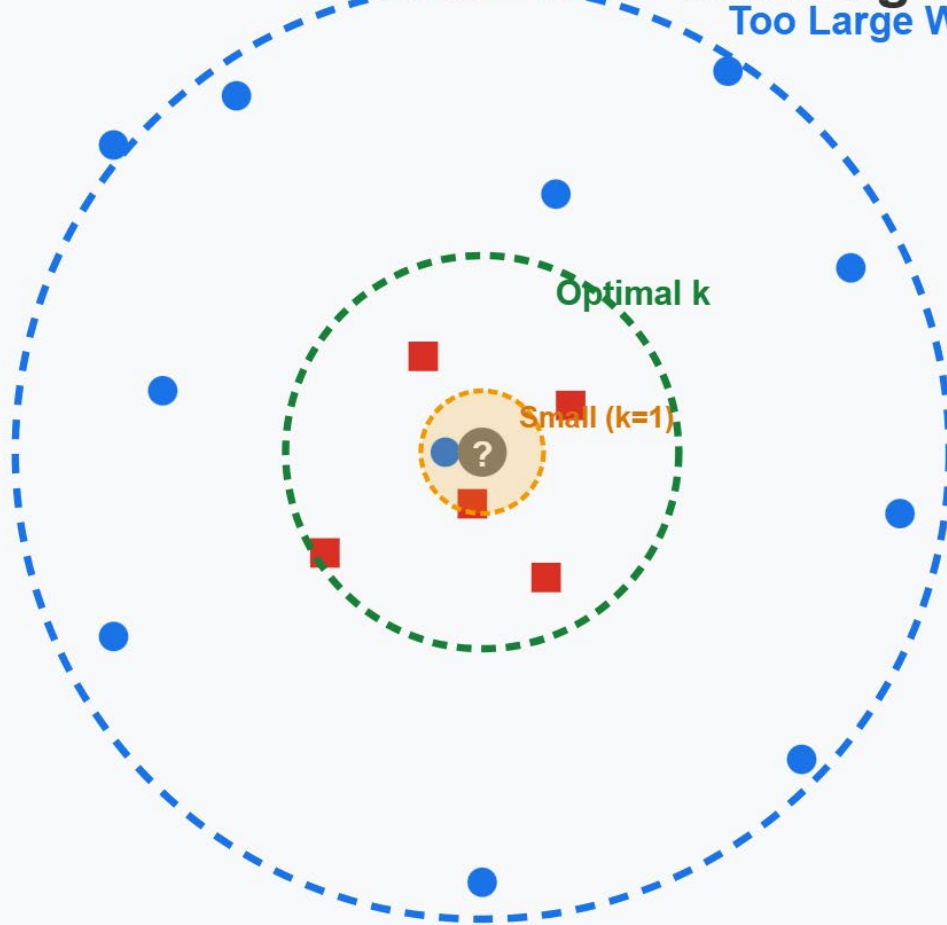
Q4. kNN Window / Neighbourhood Interpretation (Short Answer)

The lecture describes a cell centred at _____ that grows until it captures _____ samples.

1. What are those samples called?
2. If the window is too small, what problem may occur?
3. If the window is too large, what problem may occur?

(Answer each in one short phrase.)

kNN: The "Growing Window" Interpretation



1. Captured Samples

Called the **k-nearest neighbors**.

2. If window is TOO SMALL (e.g., $k=1$)

Result: **Overfitting**

It only captures the nearest point, even if it's a noise/anomaly (like the 1 blue dot inside the orange circle).

3. If window is TOO LARGE

Result: **Underfitting**

It captures too many points from other classes, over-smoothing the local boundary. The majority class always wins.

The lecture describes a cell centred at x that grows until it captures k_n samples.

1. They are called the k_n **nearest neighbours**.
2. If the window is too small: **most of the volume may be empty / over-sensitive to noise**.
3. If the window is too large: **important variations may be lost**.

Q5. kNN Computation

Use the training set below (from the lecture), where each row is

(Outlook, Temperature, Humidity, Windy, Play)

#	Outlook	Temperature	Humidity	Windy	Play
1	sunny	85	85	1.0	maybe
2	sunny	80	90	3.0	no
3	overcast	83	86	0.8	yes
4	rainy	70	96	0.2	maybe
5	rainy	68	80	0.1	yes
6	rainy	65	70	2.8	no
7	overcast	64	65	2.6	yes
8	sunny	72	95	0.3	no
9	sunny	69	70	0.5	yes
10	rainy	75	80	0.4	maybe
11	sunny	75	70	2.2	yes
12	overcast	72	90	2.4	maybe
13	overcast	81	75	0.0	yes
14	rainy	71	91	2.9	no

We want to predict Play for the case (sunny, 74, 82) as in the lecture prompt.

Assumption (to make the exercise well-defined): use Euclidean distance on (Temperature, Humidity) only, and use $k = 3$.

1. Compute the distances from (74, 82) to each training point (Temperature, Humidity).
2. Identify the 3 nearest neighbours (report their row numbers).
3. Predict the label by majority vote.

Use Euclidean distance on (Temperature, Humidity) only for the query (74, 82).

$$d = \sqrt{(T - 74)^2 + (H - 82)^2}$$

Row	(T, H)	Distance to (74, 82)	Play
1	(85,85)	$\sqrt{130} \approx 11.402$	maybe
2	(80,90)	$\sqrt{100} = 10.000$	no
3	(83,86)	$\sqrt{97} \approx 9.849$	yes
4	(70,96)	$\sqrt{212} \approx 14.560$	maybe
5	(68,80)	$\sqrt{40} \approx 6.325$	yes
6	(65,70)	$\sqrt{225} = 15.000$	no
7	(64,65)	$\sqrt{389} \approx 19.723$	yes
8	(72,95)	$\sqrt{173} \approx 13.153$	no
9	(69,70)	$\sqrt{169} = 13.000$	yes
10	(75,80)	$\sqrt{5} \approx 2.236$	maybe
11	(75,70)	$\sqrt{145} \approx 12.042$	yes
12	(72,90)	$\sqrt{68} \approx 8.246$	maybe
13	(81,75)	$\sqrt{98} \approx 9.899$	yes
14	(71,91)	$\sqrt{90} \approx 9.487$	no

1. Distances are listed in the table above.
2. The 3 nearest neighbours are rows **10, 5, 12**.
3. Their labels are maybe, yes, maybe, so the majority vote gives **maybe**.

Q6. Sensitivity to k (Structured Decision)

Using your neighbour ranking from Q5, answer:

Tie-breaking rule: if there is a tie in votes, choose the class of the closest neighbour.

1. Predicted label for $k = 3$: _____

2. Predicted label for $k = 5$: _____

3. Predicted label for $k = 7$: _____

Row 10 ($D^2 = 5$) $\rightarrow$ **maybe** (*Closest*)

Row 5 ($D^2 = 40$) $\rightarrow$ **yes**

Row 12 ($D^2 = 68$) $\rightarrow$ **maybe**

Row 14 ($D^2 = 90$) $\rightarrow$ **no**

Row 3 ($D^2 = 97$) $\rightarrow$ **yes**

Row 13 ($D^2 = 98$) $\rightarrow$ **yes**

Row 2 ($D^2 = 100$) $\rightarrow$ **no**

Using the neighbour ranking:

10, 5, 12, 14, 3, 13, 2, 1, 11, 9, 8, 4, 6, 7

1. Predicted label for $k = 3$: **maybe**
2. Predicted label for $k = 5$: **maybe**
3. Predicted label for $k = 7$: **yes**

Q7. Decision Tree Stopping Conditions (Multiple Choice)

A node can be treated as a leaf node (stop splitting) based on lecture-listed checks such as:

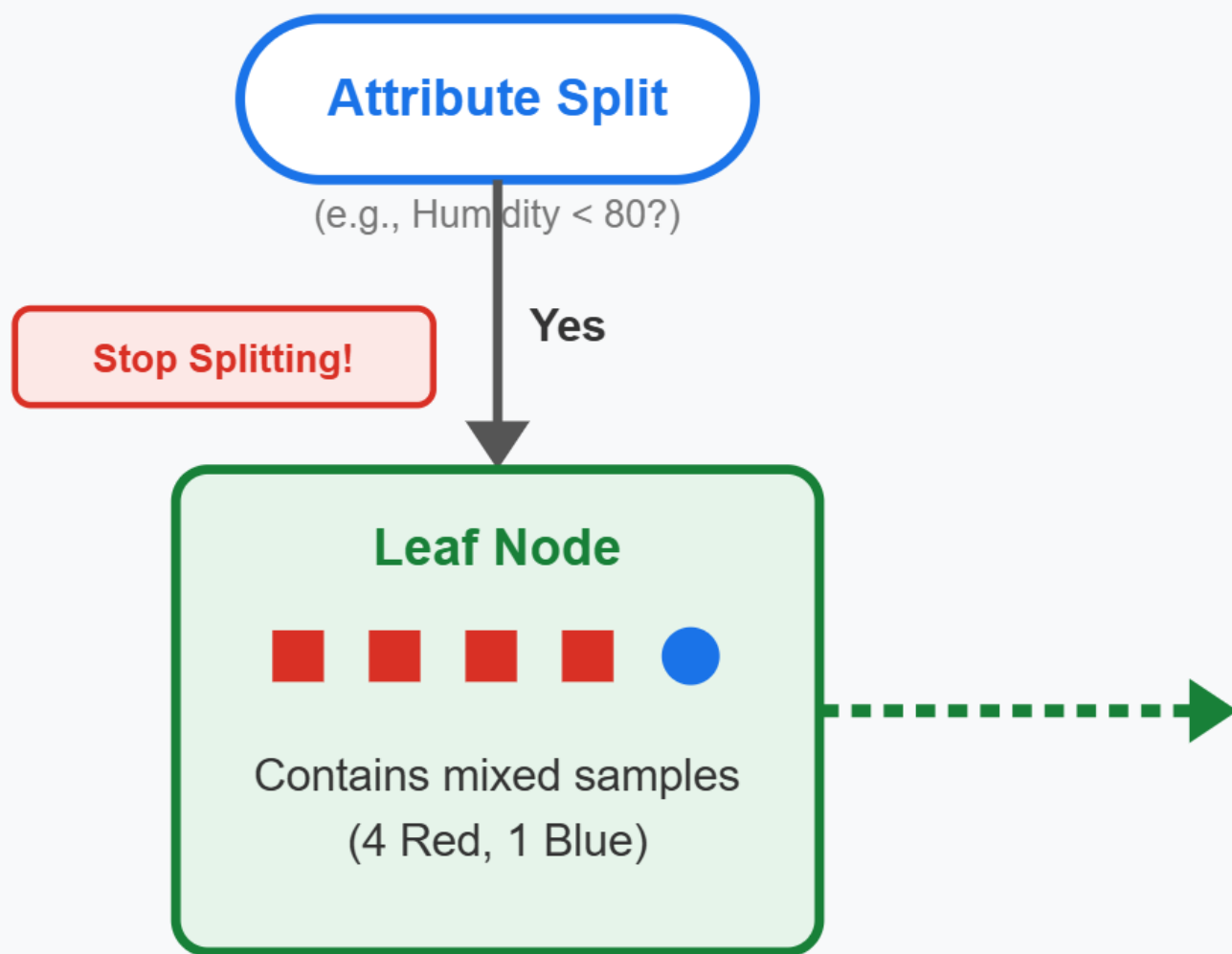
- contains no samples,
- samples belong to a universal class,
- a common attribute is shared,
- whether attributes are yet to be further analysed.

Which option best describes how the leaf decision is made?

- A. minimize Euclidean distance
- B. majority vote (mode) of the labels in the node
- C. maximize cluster similarity
- D. pick the attribute with smallest variance

Decision Tree: How a Leaf Node Makes a Decision

When splitting stops, the leaf predicts using a Majority Vote (Mode)



How to make the final decision?

Since we cannot split anymore,
we look at the remaining training labels:

■ **Class RED: 4 votes**

● Class BLUE: 1 vote

Majority Vote (Mode)
Final Prediction = RED

- **A. minimize Euclidean distance:** This is a concept used in k-Nearest Neighbors (kNN) or K-Means clustering, not in standard Decision Trees.
- **C. maximize cluster similarity:** This is a concept for unsupervised learning/clustering, not for deciding the output of a classification tree leaf.
- **D. pick the attribute with smallest variance:** Variance reduction is used *during* the splitting process (to choose which feature to split on in a Regression Tree), not for making the final prediction at the leaf.

Q8. Entropy (Fill in the blanks)

Complete the two formulas (log base 2):

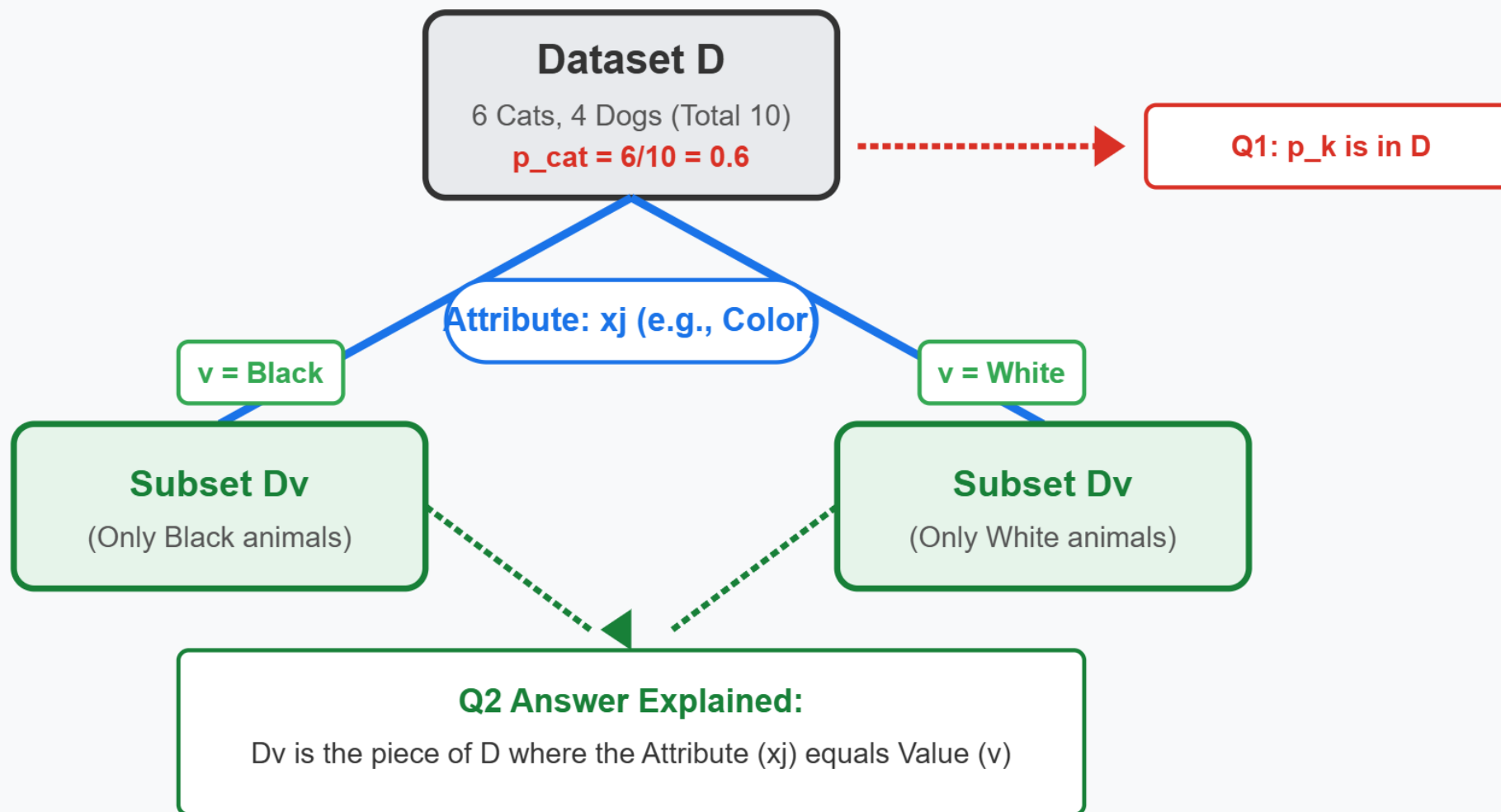
$$\text{entropy}(D) = - \sum_{k=1}^{|Y|} p_k \log_2 p_k$$

$$\text{Gain}(D, x_j) = \text{entropy}(D) - \sum_{v=1}^V \frac{|D_v|}{|D|} \text{entropy}(D_v)$$

Fill in the missing meanings:

1. p_k is the proportion of class k in _____.
2. D_v is the subset of D with attribute value _____ for _____.

Understanding Entropy & Information Gain Formulas



1. p_k is the proportion of class k in D .
2. D_v is the subset of D with attribute value v for **attribute** x_j .

Q9. Entropy Calculation (3-class Node)

A node contains 14 samples with class counts:

$$\text{Yes} = 6, \quad \text{No} = 5, \quad \text{Maybe} = 3.$$

1. Compute $\text{entropy}(D)$ (base 2), rounded to 3 decimal places.
2. State which class is the mode at this node.

Node: 14 Samples

Yes (6)



No (5)



Maybe (3)



1. Compute Entropy(D)

Formula: Entropy = $-\sum p_i \cdot \log_2(p_i)$

$$\text{Entropy}(D) = - \left[\left(\frac{6}{14}\right)\log_2\left(\frac{6}{14}\right) + \left(\frac{5}{14}\right)\log_2\left(\frac{5}{14}\right) + \left(\frac{3}{14}\right)\log_2\left(\frac{3}{14}\right) \right]$$

$$\text{Entropy}(D) = - \left[0.428 \cdot (-1.22) + 0.357 \cdot (-1.48) + 0.214 \cdot (-2.22) \right]$$

Entropy(D) $\approx$ 1.531 bits

2. State the Mode

"Mode" means the most frequent class in the node.

Max(6, 5, 3) = 6 $\rightarrow$ The mode is "Yes".

Q10. Information Gain for a Split (Computation)

Use the same parent node as Q9. Consider splitting on an attribute with three values, producing:

D_1 (size 6) : Yes = 2, No = 3, Maybe = 1

D_2 (size 4) : Yes = 3, No = 0, Maybe = 1

D_3 (size 4) : Yes = 1, No = 2, Maybe = 1

1. Compute $\text{entropy}(D_1)$, $\text{entropy}(D_2)$, $\text{entropy}(D_3)$.
2. Compute the weighted entropy after the split.
3. Compute $\text{Gain}(D, x_j)$, rounded to 3 decimals.

Parent Node D (Size: 14)

6 Yes, 5 No, 3 Maybe

Entropy(D) = 1.531

Split on Attribute X

D1 (Size: 6)

2 Yes, 3 No, 1 Maybe

E(D1) = 1.459

D2 (Size: 4)

3 Yes, 0 No, 1 Maybe

E(D2) = 0.811

D3 (Size: 4)

1 Yes, 2 No, 1 Maybe

E(D3) = 1.500

$$-\sum p_i \log_2(p_i)$$

Entropy(D₁): Has 6 samples (2 Yes, 3 No, 1 Maybe)

$$\begin{aligned} & -\left[\left(\frac{2}{6} \log_2 \frac{2}{6}\right) + \left(\frac{3}{6} \log_2 \frac{3}{6}\right) + \left(\frac{1}{6} \log_2 \frac{1}{6}\right)\right] \\ & = -[(-0.528) + (-0.500) + (-0.431)] \approx 1.459 \end{aligned}$$

Entropy(D₂): Has 4 samples (3 Yes, 0 No, 1 Maybe)

$$\begin{aligned} & -\left[\left(\frac{3}{4} \log_2 \frac{3}{4}\right) + 0 + \left(\frac{1}{4} \log_2 \frac{1}{4}\right)\right] \text{ (Note: 0 count means 0 entropy for that class)} \\ & = -[(-0.311) + 0 + (-0.500)] \approx 0.811 \end{aligned}$$

Entropy(D₃): Has 4 samples (1 Yes, 2 No, 1 Maybe)

$$\begin{aligned} & -\left[\left(\frac{1}{4} \log_2 \frac{1}{4}\right) + \left(\frac{2}{4} \log_2 \frac{2}{4}\right) + \left(\frac{1}{4} \log_2 \frac{1}{4}\right)\right] \\ & = -[(-0.500) + (-0.500) + (-0.500)] = 1.500 \end{aligned}$$

Step 2: Weighted Entropy

Weighted Avg = $\sum (\text{Size of Child} / \text{Total Size}) * E(\text{Child})$

$$= (6/14 * 1.459) + (4/14 * 0.811) + (4/14 * 1.500)$$

$$= 0.625 + 0.232 + 0.429 = 1.286$$

Step 3: Information Gain

Gain = Original Entropy - New Weighted Entropy

$$\text{Gain}(D, X_j) = 1.531 - 1.286$$

$$\text{Gain} = 0.245 \text{ bits}$$

The "Messiness" drop = The Gain!

Q11. Gain Ratio (Computation)

The lecture defines the gain ratio as

$$\text{GainRatio}(D, x_j) = \frac{\text{Gain}(D, x_j)}{\text{IV}(a)}$$

where

$$\text{IV}(a) = - \sum_{v=1}^V \frac{|D_v|}{|D|} \log_2 \left(\frac{|D_v|}{|D|} \right).$$

Using the split in Q10 (sizes 6, 4, 4 out of 14):

1. Compute IV (base 2), rounded to 3 decimals.
2. Using your $\text{Gain}(D, x_j)$ from Q10, compute $\text{GainRatio}(D, x_j)$, rounded to 3 decimals.

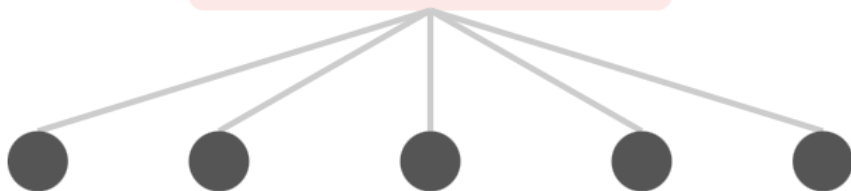
Why do we need Intrinsic Value (IV) & Gain Ratio?

Fixing the fatal flaw of Information Gain

The Flaw of Information Gain

It favors attributes that make TOO MANY splits.

Split by "Student ID"



50 Students split into 50 branches (1 person each)

Result: Max Gain, but totally useless!

Intrinsic Value (IV): The Penalty

IV measures how heavily the data is scattered.

More Branches = Higher IV

(Bigger penalty for tiny, useless pieces)

A split into 50 pieces will have a MASSIVE IV!

Gain Ratio = Information Gain / IV

By dividing by IV, we cancel out the unfair advantage of "too many splits"!

1. Calculate IV (The Penalty)

IV only looks at the sizes of the splits!
(It ignores the Yes/No labels completely).

6 / 14

Items

4 / 14

Items

4 / 14

Items

$$IV = - [(6/14)\log_2(6/14) + 2 * (4/14)\log_2(4/14)]$$

$$IV = - [(-0.524) + (-0.516) + (-0.516)]$$

$$IV = 1.557 \text{ bits}$$

Using split sizes 6, 4, 4 out of 14:

1.

$$IV = -\frac{6}{14} \log_2 \frac{6}{14} - \frac{4}{14} \log_2 \frac{4}{14} - \frac{4}{14} \log_2 \frac{4}{14} \approx 1.557$$

2.

$$\text{GainRatio}(D, x_j) = \frac{0.250}{1.557} \approx 0.161$$

2. Calculate Gain Ratio

Gain Ratio fixes the bias of Information Gain
by dividing it by the IV penalty.

$$\text{Gain Ratio} = \frac{\text{Gain (from Q10)}}{\text{IV (Penalty)}}$$

$$\text{Gain Ratio} = \frac{0.245}{1.557}$$

$$\text{Ratio} \approx 0.157$$

Q12. Clustering (Agglomerative + k-means)

Part A — Agglomerative (Hierarchical) Clustering

Using the city-block distance matrix (from the lecture example):

$$d(A, B) = 8, \quad d(A, C) = 3, \quad d(A, D) = 5, \quad d(A, E) = 13$$

$$d(B, C) = 6, \quad d(B, D) = 5, \quad d(B, E) = 10$$

$$d(C, D) = 8, \quad d(C, E) = 2$$

$$d(D, E) = 7$$

Assume single linkage (cluster distance = minimum pairwise distance).

1. What is the first merge?
2. What is the second merge?
3. List the clusters after the second merge.

Part B — One Iteration of k-means

Given 7 points (from the lecture example):

$$(1.0, 1.0), (1.5, 2.0), (3.0, 4.0), (5.0, 7.0), (3.5, 5.0), (4.5, 5.0), (3.5, 4.5)$$

Let $k = 2$ and initialise centroids as:

$$c_1^{(0)} = (1.0, 1.0), \quad c_2^{(0)} = (5.0, 7.0).$$

Use Euclidean distance. If a point is equidistant, assign it to the smaller-index cluster.

1. Assign each point to Cluster 1 or Cluster 2.
2. Compute the updated centroids $c_1^{(1)}$ and $c_2^{(1)}$.

Part A — Agglomerative (Hierarchical) Clustering

Using single linkage and the city-block distances:

1. The first merge is C **and** E , since

$$d(C, E) = 2$$

is the smallest distance.

2. The second merge is A **with** (C, E) , because

$$d(A, C) = 3, \quad d(A, E) = 13,$$

so by single linkage

$$d(A, \{C, E\}) = \min(3, 13) = 3.$$

3. The clusters after the second merge are:

$$\{A, C, E\}, \quad \{B\}, \quad \{D\}.$$

$$d(\{C, E\}, A) = \min(d(C, A), d(E, A)) = \min(3, 13) = 3$$

$$d(\{C, E\}, B) = \min(d(C, B), d(E, B)) = \min(6, 10) = 6$$

$$d(\{C, E\}, D) = \min(d(C, D), d(E, D)) = \min(8, 7) = 7$$

Part B — One Iteration of k-means

Given 7 points (from the lecture example):

$$(1.0, 1.0), (1.5, 2.0), (3.0, 4.0), (5.0, 7.0), (3.5, 5.0), (4.5, 5.0), (3.5, 4.5)$$

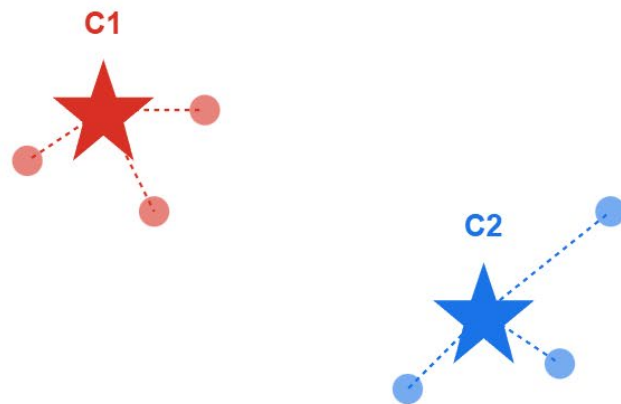
Let $k = 2$ and initialise centroids as:

$$c_1^{(0)} = (1.0, 1.0), \quad c_2^{(0)} = (5.0, 7.0).$$

Use Euclidean distance. If a point is equidistant, assign it to the smaller-index cluster.

1. Assign each point to Cluster 1 or Cluster 2.
2. Compute the updated centroids $c_1^{(1)}$ and $c_2^{(1)}$.

Step 1: Assign to Closest Centroid



Step 2: Update Centroids (Take Average)



The star moves to the exact center of its points!

Part B — One Iteration of k-means

Points:

$$(1.0, 1.0), (1.5, 2.0), (3.0, 4.0), (5.0, 7.0), (3.5, 5.0), (4.5, 5.0), (3.5, 4.5)$$

Initial centroids:

$$c_1^{(0)} = (1.0, 1.0), \quad c_2^{(0)} = (5.0, 7.0)$$

1. Cluster assignment after one iteration:

$$(1.0, 1.0) \rightarrow C_1$$

$$(1.5, 2.0) \rightarrow C_1$$

$$(3.0, 4.0) \rightarrow C_1$$

$$(5.0, 7.0) \rightarrow C_2$$

$$(3.5, 5.0) \rightarrow C_2$$

$$(4.5, 5.0) \rightarrow C_2$$

$$(3.5, 4.5) \rightarrow C_2$$

So,

$$C_1 = \{(1.0, 1.0), (1.5, 2.0), (3.0, 4.0)\}$$

$$C_2 = \{(5.0, 7.0), (3.5, 5.0), (4.5, 5.0), (3.5, 4.5)\}$$

2. Updated centroids:

$$c_1^{(1)} = \left(\frac{1.0 + 1.5 + 3.0}{3}, \frac{1.0 + 2.0 + 4.0}{3} \right) = \left(\frac{5.5}{3}, \frac{7}{3} \right) \approx (1.833, 2.333)$$

$$c_2^{(1)} = \left(\frac{5.0 + 3.5 + 4.5 + 3.5}{4}, \frac{7.0 + 5.0 + 5.0 + 4.5}{4} \right) = \left(\frac{16.5}{4}, \frac{21.5}{4} \right) = (4.125, 5.375)$$